

Carbon-optimal data-center siting is not robust to the carbon-accounting choice: a specification-curve analysis on live grid data

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Abstract

The surge in data-center load has made “where should the next data center go to minimize carbon?” a live policy question, usually answered by ranking regions on a single emission factor. **Our single claim: that ranking—and therefore the carbon-optimal siting recommendation—is not robust to the carbon-accounting choice.** Using two years of hourly EIA-930 data for 27 US balancing authorities (BAs) across all three interconnects, with the computed average factor validated against EIA’s own published hourly CO₂ to a median of 1.6% for fossil-dominated BAs, we compute a per-BA carbon factor under a *multiverse* of defensible accounting choices spanning four axes: temporal (short-run dispatch vs long-run build margin), accounting boundary (production vs consumption/import-adjusted), spatial resolution (BA vs nodal), and estimation method (regression vs supply-side instrument vs capacity-expansion model). Across six specifications, **24 of 27 BAs change siting rank and only 3 are robust** (two Pacific-Northwest hydro BAs robustly green, one coal BA robustly dirty). The re-rankings are not cosmetic: the median flipping BA moves 12 ranks, 10 BAs span at least half the fleet, and four undergo outright inversions—Idaho Power (IPCO) runs from rank 3 (near-greenest, short-run) to 25 (near-worst, long-run). The short-run marginal factor is itself hard to identify: three independent supply-side estimators converge within 20% for only 6 of 23 BAs, which we read as evidence *for* the fragility thesis—the margin is genuinely ill-defined outside a single-fossil-unit regime—not as validation. We turn the critique into a decision rule: site where the verdict is robust across all accounting choices (the Pacific-NW hydro core), or, where that core is unavailable, hedge across regions whose rank vectors are anti-correlated (IPCO+MISO, rank-correlation -0.93 , caps worst-case rank at 9 versus 25 for either site alone). All numbers regenerate from the released **gridpulse** pipeline.

1 Introduction

A new data center is an increment of load that persists for 15+ years. Its climate impact is therefore governed by the grid’s **marginal** response, not its average—a point established since Siler-Evans et al. (2012). The policy literature and commercial siting tools translate this into a ranking: order BAs by a carbon factor, site in the low-carbon ones. **We show that ranking is not robust.** “The marginal factor” hides four choices, each contested at the research frontier and each capable of inverting which region is “greenest”:

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1. **Temporal horizon.** The short-run *dispatch* margin (which existing unit ramps) differs from the long-run *build* margin (what new capacity the load induces). Short-run-optimal operation can raise long-run emissions (Gagnon & Cole 2022); marginal factors have not fallen as fast as average factors (Holland et al. 2022).
2. **Accounting boundary.** Production-based emissions ignore that a BA may serve load with dirty imports; consumption-based (import-adjusted) accounting via interchange flow-tracing (de Chalendar et al. 2019) re-ranks regions.
3. **Spatial resolution.** Two nodes in the same BA face different marginal carbon under transmission congestion; BA-average erases this.
4. **Estimation method.** Regression (Siler-Evans), dispatch/instrumental estimates, and capacity-expansion models (NREL Cambium) need not agree.

Our contribution is a single, sharp claim answered with a specification-curve (multiverse) analysis (Simonsohn, Simmons & Nelson 2020): **a carbon-optimal siting recommendation is meaningless without the accounting choice that produced it, and the set of sites that are good under *every* choice is small.** We build and validate each method on live data, release the full pipeline as **gridpulse**, and close with a decision rule that follows directly from the fragility we measure.

2 Data

- **EIA-930 hourly** fuel mix, demand, and BA-to-BA interchange for 27 balancing authorities across all three interconnects, via the EIA API v2. 24 months at full coverage; 2019–present for a subset (Holland test). These 27 are a **truncation of the ~66-BA EIA interchange network** (see §6).
- **EIA Hourly Electric Grid Monitor** per-BA workbooks—EIA’s *published* hourly CO₂, used as validation ground truth (not self-consistency).
- **NREL Cambium** long-run marginal emission rates (LRMER), GEA-region resolution, forward scenarios.
- **CAISO OASIS** nodal LMP with congestion decomposition (no-auth, reproducible), for the intra-BA nodal analysis.
- **FracTracker** national data-center tracker (per-facility MW + location).

3 Methods

- **AEF:** $\sum CO_2 / \sum \text{generation}$, on EIA’s production basis (fossil combustion only: COL, NG, OIL), validated against EIA’s published hourly CO₂.
- **Short-run MEF:** Siler-Evans OLS of ΔCO_2 on Δdemand over consecutive hours, bootstrap CIs; recovers a planted margin to machine precision on synthetic data.
- **Consumption MEF:** emissions flow-tracing over the interchange network (de Chalendar 2019 linear system), then a marginal factor on the import-adjusted series. Runs on the 27-BA modeled subnetwork; imports across cut edges are dropped (§6).

- **Supply-side instruments** (identification stress-test): (i) VRE-ramp—carbon displaced per MWh of exogenous wind/solar with demand held flat; (ii) generation-trip—emissions response to detected forced outages.
- **Nodal LME**: CAISO nodal congestion component modulating the BA system margin (directional only; exact nodal kg/MWh requires proprietary CEII shift factors).
- **Long-run**: NREL Cambium LRMER/SRMER, mapped GEA-region $\rightarrow$ BA.
- **Specification curve**: rank BAs under every (temporal $\times$ accounting $\times$ spatial $\times$ method) combination; classify each BA robust-green / robust-dirty / flips by its rank range across specifications.

4 Results

Generated by the live run (`gridpulse phase3`); tables in `FINDINGS.md/EVIDENCE.md`, committed rank matrix in `data/multiverse_ranks.csv` and robustness classification in `data/multiverse_robustness.csv`.

4.1 Validation (of the pipeline, not the thesis). Computed AEF matches EIA’s *published* hourly CO₂ to a median of **1.6%** across fossil-dominated BAs using EIA’s own per-fuel factors (ERCO -2.4% , MISO -0.6% , DUK -5.4% , SOCO -1.5% ; Figure 1). This anchors the average-factor pipeline before any marginal analysis; it says nothing about how well-identified the *marginal* factor is (see 4.2).

4.2 The short-run margin is hard to identify—evidence for fragility. Siler-Evans, the VRE-ramp instrument, and generation-trip events converge within 20% for **only 6 of 23 BAs** (FPC, PJM, SWPP, SOCO, MISO, AZPS)—exactly those where a single fossil unit sets the margin. In storage/solar/import-heavy systems the three estimates diverge by 100–800% (CISO 141%, WACM 573%, NEVP 847%), because batteries and imports break the one-margin assumption. We do **not** read 6/23 as validation of the MEF. It is direct evidence that the object the siting literature ranks on is genuinely ill-defined for three-quarters of the fleet—a first-order reason the downstream ranking is fragile.

4.3 Consumption re-ranking. Import-adjusted (flow-traced) marginal carbon moves several BAs 10+ ranks versus production-based (CPLE -14 , PGE $+11$, ISNE -10).

4.4 Nodal. Within CISO, congestion spreads the directional marginal-carbon index $\pm\sim 50\%$ around the BA-average (p10–p90 ≈ 89 – 202 kg/MWh vs 142 system); cleanest nodes are export-constrained **solar** buses, dirtiest are import-constrained load pockets—variation BA-resolution erases entirely.

4.5 Short-run vs long-run. Ranking under NREL Cambium **LRMER** (long-run) vs our short-run MEF flips hydro-rich BAs hardest: **IPCO shifts -22 ranks** (rank 3 short-run—its extra MWh is existing hydro; rank 25 long-run—new load induces fossil capacity), MISO $+16$, ISNE $+12$, PACE -11 , FPL -9 . Separately, reproducing Holland et al. (2022) on 2019–2025 data, **4 of 8 BAs show the marginal factor falling more slowly than the average** (stickiness < 1): renewables cut the average while gas still sets the margin (Figure 3).

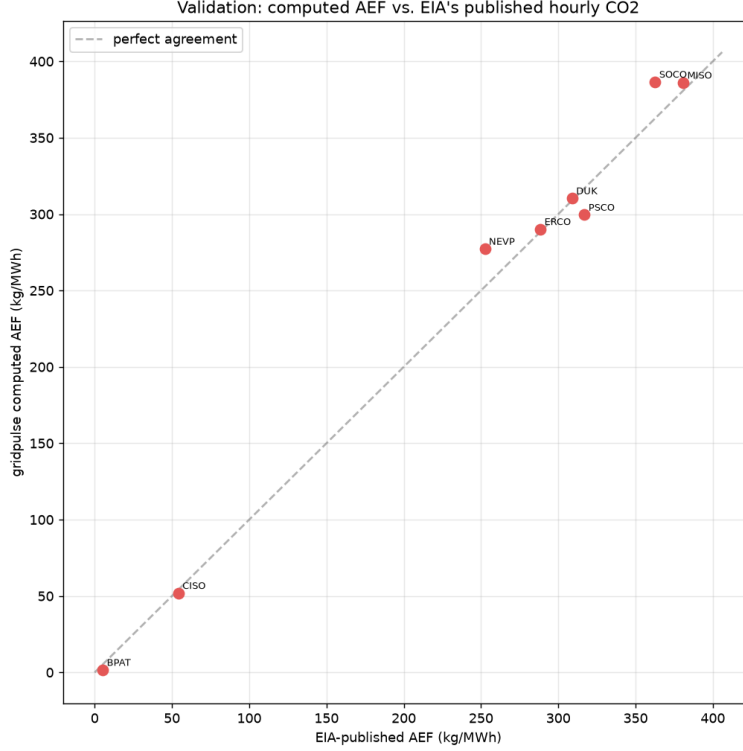


Figure 1: Computed AEF vs. EIA’s published hourly CO₂ for fossil-dominated BAs. Points hug the identity line (median deviation 1.6%), anchoring the average-factor pipeline before any marginal analysis.

4.6 Multiverse (headline). Across six specifications—AEF; short-run production MEF (regression); short-run production MEF (VRE instrument); short-run consumption MEF (flow-trace); long-run LRMER (Cambium); short-run SRMER (Cambium)—**24 of 27 BAs change siting rank; only 3 are robust:** BPAT (rank range 1–2) and PACW (1–3) robustly green, LGEE (24–27) robustly dirty (Figure 4). The distribution of the 24 flips by rank range (max – min across specs; median 12 of a 27-BA fleet) is:

flip magnitude		<i>n</i>	BAs (rank range)
minor	(≤5 ranks)	1	NEVP 5
moderate	(6–10)	9	NYIS 10, CISO 9, PSCO 9, SRP 9, PJM 9, SWPP 8, ERCO 7, TVA 7, SOCO 6
large	(11–15)	10	AZPS 15, WACM 15, PGE 14, CPLE 14, FPL 14, LDWP 13, ISNE 12, DUK 12, FPC 12, PNM 11
inversion	(>15)	4	IPCO 22 (3→25) , PACE 19 (5→24), AECI 19 (7→26), MISO 16 (9→25)

Table 1: Distribution of the 24 flipping BAs by rank range across the six specifications. 10 of the 24 span at least half the 27-BA fleet (range ≥13); four are outright inversions.

So the headline is not a uniform reshuffle: 10 of the 24 flippers span **at least half the fleet** (range ≥13), and four are outright inversions where a BA moves from the greenest quartile to the dirtiest (or vice versa) on the accounting choice alone. Only the Pacific-NW hydro core is unambiguous.

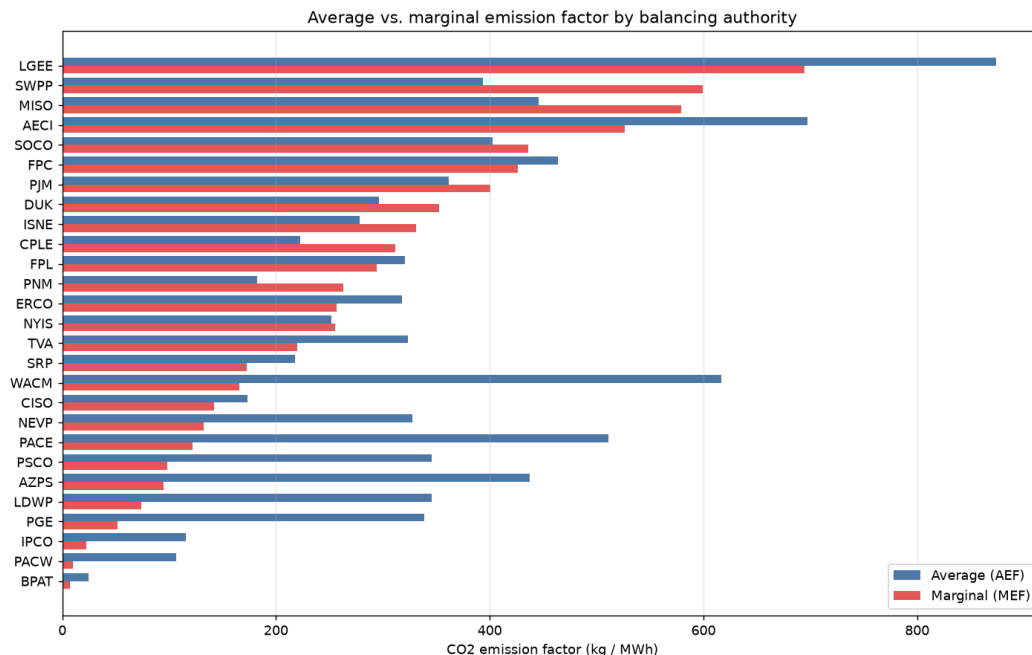


Figure 2: Average (AEF) vs. short-run marginal (MEF) emission factor by BA. The two orderings are visibly different—BAs clean on average are not the same set as BAs clean on the margin—which is the raw material for the re-rankings in §4.6.

5 Discussion

The carbon-optimal siting recommendation is contingent on the analyst’s accounting choices to a degree that undermines single-metric guidance. The robust policy signal is narrow—Pacific-NW hydro—and everywhere else the verdict depends on whether one prices short- or long-run carbon, production or consumption, BA-average or nodal, and by which estimator. Siting policy and 24/7-CFE procurement (Ricks, Xu & Jenkins 2023) should therefore report the accounting choice explicitly and prefer regions robust across the multiverse. §5.1 makes this actionable.

5.1 A decision rule under fragility

Fragility is not a dead end for a developer; it prescribes two concrete strategies, both computed from the committed multiverse output.

Rule A—site where the verdict is robust. Only BPAT and PACW are green under *every* one of the six accounting choices (rank ranges 1–2 and 1–3). A recommendation to build there carries no accounting caveat. Concentrating a carbon-optimal 10 GW build-out in the low-carbon set—respecting a 20%-of-local-demand deliverability cap per BA—avoids $\approx 25.2 \text{ MtCO}_2/\text{yr}$ (76% of the tracked build-out’s marginal emissions) on the production-marginal basis (best site BPAT at 7.4 kg/MWh; `data/gap.json`). This payoff is itself accounting-dependent— $31.7 \text{ MtCO}_2/\text{yr}$ on the consumption-marginal basis—which is the point: even the *size of the prize* moves with the accounting choice, so the robust core is where a developer captures it without betting on which choice is right.

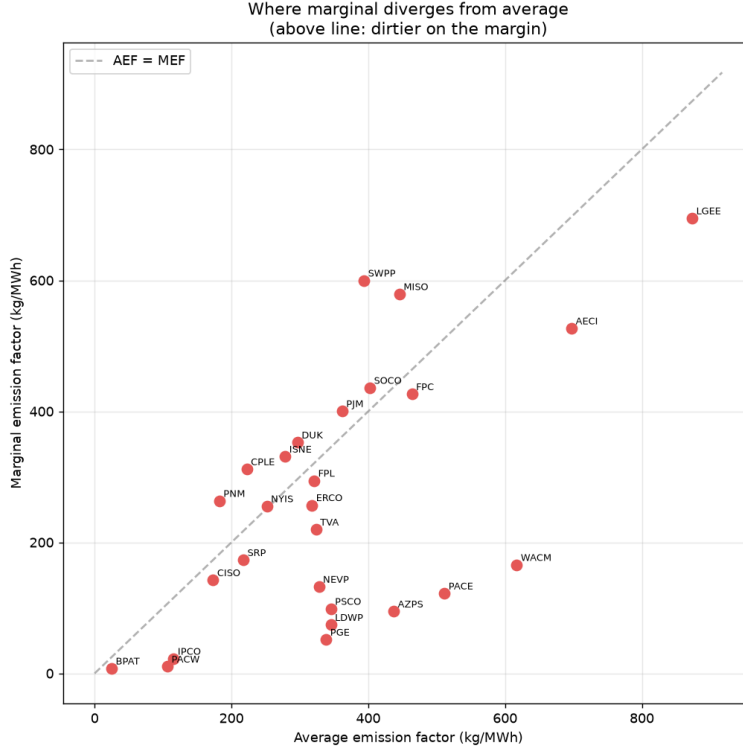


Figure 3: Where marginal diverges from average. BAs above the diagonal are dirtier on the margin than on average; the ordering along either axis is not preserved on the other, so which BA is “greenest” depends on which factor is priced.

Rule B—hedge across anti-correlated regions. Where the hydro core is unavailable (deliverability, land, water, latency), split load across a pair of BAs whose rank vectors across the six specifications are anti-correlated, so the portfolio is insensitive to which accounting choice ultimately governs. The cleanest case is the temporal axis: **IPCO** (rank 3–6 short-run, 25 long-run) and **MISO** (25 short-run, 9 long-run) have a rank-vector correlation of -0.93 ; a developer able to take the better of the two under whichever horizon prices the carbon is **never worse than rank 9**, versus rank 25 for a single-site commitment. On the production-vs-consumption axis, **PNM+WACM** (correlation -0.96) cap the worst-case rank at 12 versus 25 for WACM alone. (This assumes load can shift between the paired sites and that the measured anti-correlation persists out of sample; the correlation is computed over the six specifications, not forecast.)

6 Limitations

- **Subnetwork truncation (primary exposure).** All results hold **within the 27-BA modeled subnetwork**, a truncation of the ~ 66 -BA EIA interchange network. Flow-tracing (§3) is solved on this subnetwork, so edges to unmodeled BAs are cut and imports across them are dropped. This is why the consumption MEF is **floored at 0** for solar-saturated, strongly import-dependent BAs (CISO): with the import edges removed, the raw import-adjusted estimate dips physically-impossibly negative and is clipped to the non-negative minimum. **Direction of bias:** the floor makes such BAs look artificially clean on the consumption specification, which *widens* their measured rank range (CISO reaches rank 1 on the consumption

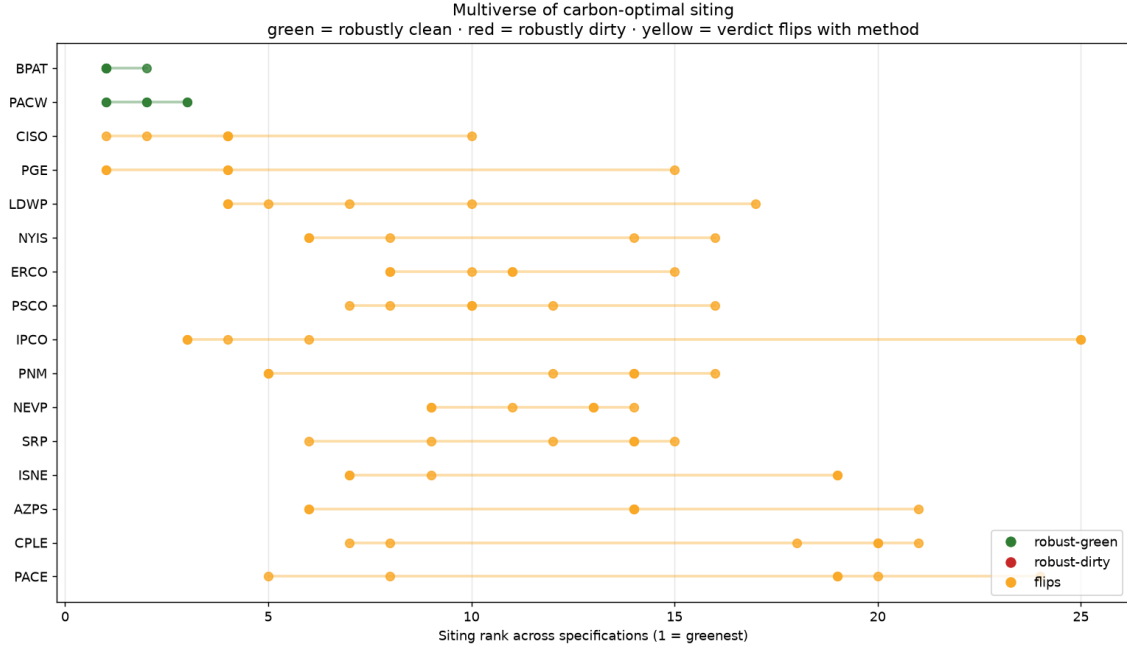


Figure 4: Specification-curve (multiverse) of carbon-optimal siting rank. Each row is a BA; the horizontal spread is its siting rank across the six accounting specifications (1 = greenest). Green = robust-green (BPAT, PACW), red = robust-dirty (LCEE), yellow = verdict flips with the accounting choice. Only 3 of 27 BAs are robust.

spec vs rank 10 on regression MEF). The truncation therefore, if anything, **overstates the flip magnitude for import-dependent BAs**; restoring the omitted (largely fossil) out-of-subnetwork imports would raise their consumption factors and *narrow* those particular flips. It does **not** manufacture the flip—CISO already re-ranks on the production bases alone—and it does **not** touch the robust core: BPAT and PACW are net hydro *exporters*, not import-sensitive. So the central claim (a small robust set; most BAs contingent) is conservative to this bias, while specific solar-BA flip magnitudes should be read as upper bounds. A full ~66-BA solve is the natural next step.

- **Short-run vs long-run dependence.** Our short-run MEF is a dispatch-margin estimate; Cambium LRMER is the long-run build margin. The IPCO 3→25 inversion is driven entirely by this axis, so any headline flip involving it inherits the short-run/long-run modeling assumptions and is not a single ground-truth number.
- **Cambium scenario dependence.** The long-run results use Cambium’s Mid-case, 2025–2030 vintage. LRMER/SRMER are model- and scenario-dependent; a different scenario (high-renewable, high-gas) would move the long-run ranks and could change which BAs invert.
- **Nodal estimates are directional only.** The nodal LME is a congestion-based *index*; an exact nodal kg/MWh needs proprietary/CEII shift factors and offer stacks that are not public. We report spread and direction, not levels, and there is **no nodal ground truth** to validate against.
- **MEF identification** degrades in storage/solar/import-heavy BAs (§4.2; shown, not hidden)—the short-run margin is well-identified in only 6 of 23 BAs.

- Two-year primary window; longer history (2019–present) only for the 8-BA subset used in the Holland test.

7 Reproducibility

All results regenerate from zero via `gridpulse` (`pip install -e ".[prod]"; pytest`; the CLI + `phase3` orchestrator). Data sources are public; the only credential is a free EIA API key. CI runs the live refresh monthly. Cite via `CITATION.cff` (Zenodo DOI on release).

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